i.e. we compose the flows of

$$\dot{\boldsymbol{\pi}} = \boldsymbol{\pi} \times \begin{bmatrix} T_1^{-1}\pi_1 \\ 0 \\ 0 \end{bmatrix}, \qquad \dot{\boldsymbol{\pi}} = \boldsymbol{\pi} \times \begin{bmatrix} 0 \\ T_2^{-1}\pi \\ 0 \end{bmatrix}, \qquad \dot{\boldsymbol{\pi}} = \boldsymbol{\pi} \times \begin{bmatrix} 0 \\ 0 \\ T_3^{-1}\pi \end{bmatrix},$$

Each of these is, it turns out, easily integrated as a planar rotation (see Exercise 7). This yields a simple, efficient algorithm for the free rigid body which is symplectic in a generalized sense.

4.4.2 Rigid Body Molecular Dynamics Algorithm

To develop an integrator for the complete system (including external forces), we invoke the splitting technique,[2] solving successively for the translational motion from the center of mass equations, integrating the torques due to the rotational interactions, then solving the Euler equations, and recovering the rotation matrix from the linear system

$$\dot{\boldsymbol{\Theta}} = \boldsymbol{\Theta}\, \mathrm{skew}(\boldsymbol{T}^{-1}\boldsymbol{\pi}).$$

This is the approach described in [112] and is implemented in several software packages.

The reliability of these methods is supported by the fact that the combined scheme is a symplectic method in the generalized sense of constrained dynamics.

The specific steps of the algorithm proposed in [112] are given here. In what follows, we denote the variables associated to the kth rigid body by $\boldsymbol{q}^k$ (position of center of mass), $\boldsymbol{\Theta}^k$ (3×3 rotation matrix), $\boldsymbol{p}^k$ (linear momentum vector), and $\boldsymbol{\pi}^k$ (angular momentum vector). Individual components are defined by double subscripts: the 2nd component of $\boldsymbol{\pi}^k$ is π_2^k. Procedures given below for rigid body k are meant to be performed for all rigid bodies in the system.

The algorithm is initialized by computing the forces $\boldsymbol{F}^k$ and torques $\boldsymbol{\tau}^k$ acting on the kth rigid body, evaluated at the initial point. The relevant expressions are:

$$\boldsymbol{F}^k = -\frac{\partial U}{\partial \boldsymbol{q}^k}, \qquad \boldsymbol{\tau}^k = -\mathrm{rot}\left(\left[\boldsymbol{\Theta}^k\right]^T \frac{\partial U}{\partial \boldsymbol{\Theta}^k} \right).$$

[2] We note the seminal role of the work in astronomy of Touma and Wisdom [368] in elucidating the "divide and conquer" approach for deriving such schemes for rigid bodies.

The notation $\partial U/\partial \boldsymbol{\Theta}^k$ refers to a 3×3 matrix of partial derivatives of the potential with respect to the corresponding elements of the rotation matrix $\boldsymbol{\Theta}^k$, and, for a 3×3 matrix $\boldsymbol{A}$,

$$\mathrm{rot}(\boldsymbol{A}) = \mathrm{skew}^{-1}(\boldsymbol{A} - \boldsymbol{A}^T).$$

Then the algorithm proceeds by iteration of timesteps, as shown below.

DLM Rigid Body Molecular Dynamics Step

1. "Kick": update the momenta and angular momentum vectors with known forces and torques for a half step:

$$\boldsymbol{p}^k := \boldsymbol{p}^k + \frac{h}{2}\boldsymbol{F}^k, \qquad \boldsymbol{\pi}^k := \boldsymbol{\pi}^k + \frac{h}{2}\boldsymbol{\tau}^k.$$

2. "Drift": step forward a whole step in positions

$$\boldsymbol{q}^k := \boldsymbol{q}^k + h\boldsymbol{p}^k.$$

3. "Spin": update the rotational state of the kth rigid body by the sequence of steps

$$\begin{aligned}
&\boldsymbol{\Theta} := \boldsymbol{\Theta}^x\left(\tfrac{h}{2}\pi_1^k/I_1\right);\ \boldsymbol{\pi}^k := \boldsymbol{\Theta}\boldsymbol{\pi}^k;\ \boldsymbol{\Theta}^k := \boldsymbol{\Theta}^k\boldsymbol{\Theta}^T\\
&\boldsymbol{\Theta} := \boldsymbol{\Theta}^y\left(\tfrac{h}{2}\pi_2^k/I_2\right);\ \boldsymbol{\pi}^k := \boldsymbol{\Theta}\boldsymbol{\pi}^k;\ \boldsymbol{\Theta}^k := \boldsymbol{\Theta}^k\boldsymbol{\Theta}^T\\
&\boldsymbol{\Theta} := \boldsymbol{\Theta}^z\left(h\pi_3^k/I_3\right);\ \boldsymbol{\pi}^k := \boldsymbol{\Theta}\boldsymbol{\pi}^k;\ \boldsymbol{\Theta}^k := \boldsymbol{\Theta}^k\boldsymbol{\Theta}^T\\
&\boldsymbol{\Theta} := \boldsymbol{\Theta}^y\left(\tfrac{h}{2}\pi_2^k/I_2\right);\ \boldsymbol{\pi}^k := \boldsymbol{\Theta}\boldsymbol{\pi}^k;\ \boldsymbol{\Theta}^k := \boldsymbol{\Theta}^k\boldsymbol{\Theta}^T\\
&\boldsymbol{\Theta} := \boldsymbol{\Theta}^x\left(\tfrac{h}{2}\pi_1^k/I_1\right);\ \boldsymbol{\pi}^k := \boldsymbol{\Theta}\boldsymbol{\pi}^k;\ \boldsymbol{\Theta}^k := \boldsymbol{\Theta}^k\boldsymbol{\Theta}^T
\end{aligned}$$

   The notation $\boldsymbol{\Theta}^x(\phi)$ refers to a rotation around the x-axis by an angle ϕ, that is

$$\boldsymbol{\Theta}^x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix},$$

   with corresponding formulae for $\boldsymbol{\Theta}^y$ and $\boldsymbol{\Theta}^z$.
4. "Kick": update the momenta and angular momentum vectors (as in Step 1) with a half step using new forces $\boldsymbol{F}^k$ and torques $\boldsymbol{\tau}^k$ evaluated at the current $\boldsymbol{q}$, $\boldsymbol{\Theta}$.

Note that the symmetric character of the algorithm could be revealed more clearly by dividing the "Drift" step into two half steps sandwiched around the Spin step (Kick, Drift, Spin, Drift, Kick), but as the updates Drift/Spin are decoupled, the order they are performed is not important.